

Statistical Analysis Report

Analysis Summary Template

Ed Harris

2025-10-20

Table of contents

1	Summary	2
2	Introduction	2
3	Data	2
4	Exploratory Data Analysis	3
4.1	Univariate Analysis	3
4.1.1	Numerical Variables	3
4.1.2	Categorical Variables	4
4.2	Bivariate Analysis	6
4.2.1	Relationships Between Variables	6
4.2.2	Correlation Analysis	7
5	Statistical Methods	8
5.1	Methodology	8
5.2	Assumptions	8
5.2.1	Checking Assumptions	8
6	Results	8
6.1	Primary Analysis	8
6.1.1	Model Fitting	8
6.1.2	Model Diagnostics	9
6.1.3	Results Summary	10
6.1.4	Interpretation	11
6.2	Secondary Analyses	11
6.2.1	Additional Tests	11
6.2.2	Subgroup Analysis	11

6.3	Effect Sizes and Practical Significance	12
7	Discussion	12
7.1	Key Findings	12
7.2	Limitations	12
7.3	Implications	12
7.4	Future Directions	13
8	Conclusions	13
9	References	13
10	Appendix	13
10.1	Session Information	13
10.2	Additional Tables and Figures	14

1 Summary

Provide a brief overview of the analysis objectives, key findings, and main conclusions. This section should be concise and accessible to non-technical stakeholders. Max 1 paragraph, no bullet points

2 Introduction

Briefly describe the context and motivation for the analysis. What problem are you trying to solve or what question are you trying to answer? List the main objectives of the analysis. Max 1 paragraph, no bullet points.

3 Data

Describe where the data comes from, how it was collected, and any relevant details about data provenance. Max 1 paragraph (usually a few sentences unless the experimental design is described in detail), no bullet points.

```
# Load your data
# data <- read.csv("your_data.csv")

# For demonstration, using built-in dataset
data <- mtcars
```

```
# Display structure of the data
str(data)
```

```
'data.frame': 32 obs. of 11 variables:
 $ mpg : num 21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num 6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num 160 160 108 258 360 ...
 $ hp : num 110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num 3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt : num 2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num 16.5 17 18.6 19.4 17 ...
 $ vs : num 0 0 1 1 0 1 0 1 1 1 ...
 $ am : num 1 1 1 0 0 0 0 0 0 0 ...
 $ gear: num 4 4 4 3 3 3 3 4 4 4 ...
 $ carb: num 4 4 1 1 2 1 4 2 2 4 ...
```

```
# Display first few rows
head(data) %>%
  kable(caption = "Dataset sample")
```

Table 1: Dataset sample

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Sportabout											
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

4 Exploratory Data Analysis

4.1 Univariate Analysis

4.1.1 Numerical Variables

```
# Example: Histogram
data %>%
  ggplot(aes(x = mpg)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  labs(title = "Distribution of Miles per Gallon",
       x = "MPG",
       y = "Frequency") +
  theme_minimal()
```

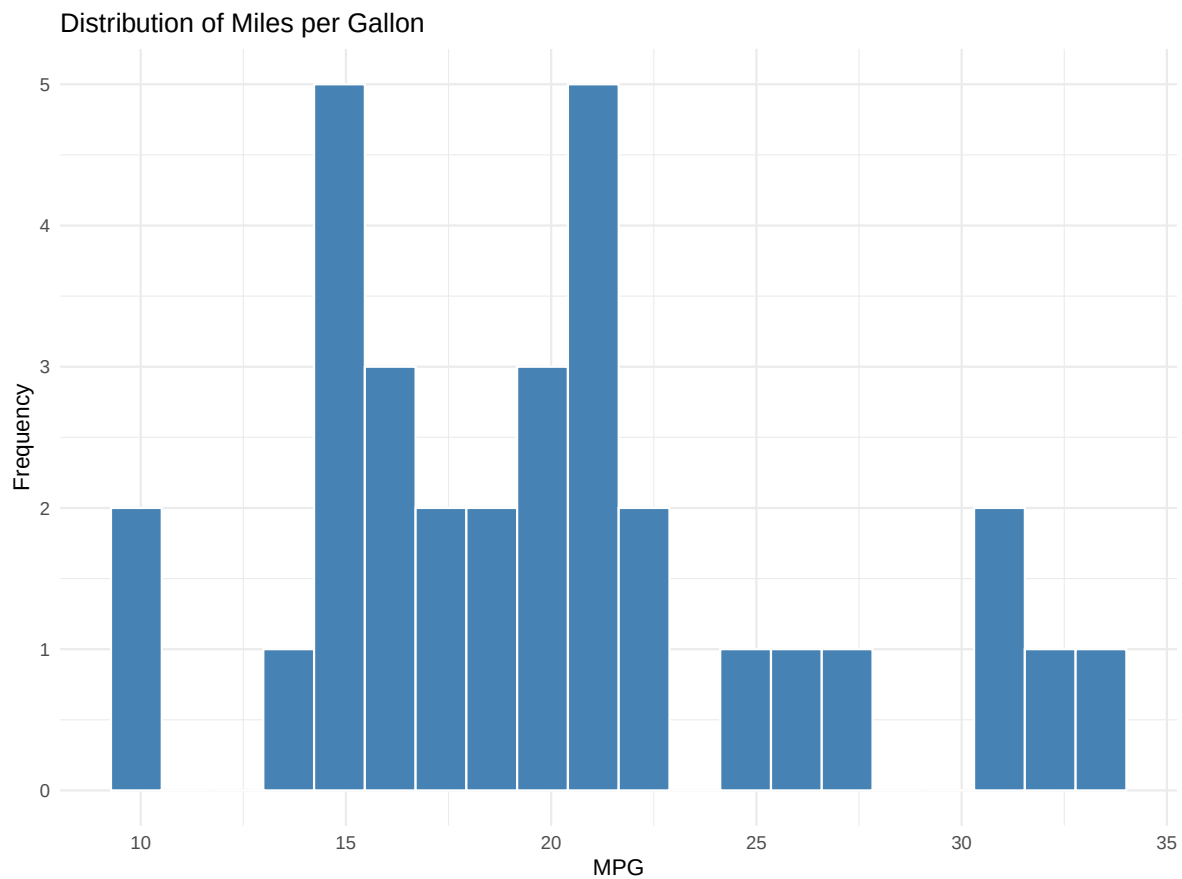


Figure 1: Distribution of numerical variables

4.1.2 Categorical Variables

```
# Example: Bar plot
data %>%
  ggplot(aes(x = factor(cyl))) +
  geom_bar(fill = "coral") +
  labs(title = "Distribution of Cylinders",
       x = "Number of Cylinders",
       y = "Count") +
  theme_minimal()
```

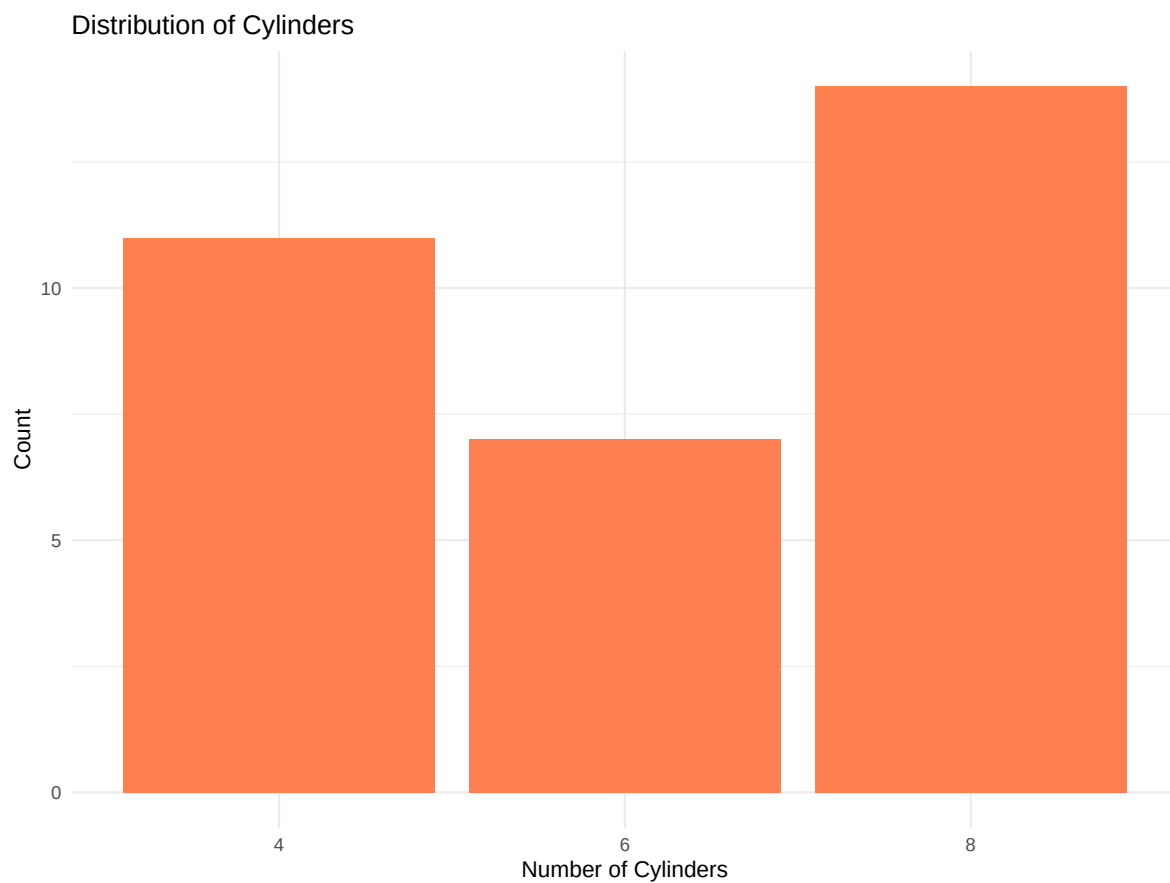


Figure 2: Distribution of categorical variables

4.2 Bivariate Analysis

4.2.1 Relationships Between Variables

```
# Example: Scatter plot
data %>%
  ggplot(aes(x = wt, y = mpg)) +
  geom_point(size = 3, alpha = 0.6, color = "darkblue") +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  labs(title = "Relationship between Weight and MPG",
       x = "Weight (1000 lbs)",
       y = "Miles per Gallon") +
  theme_minimal()
```

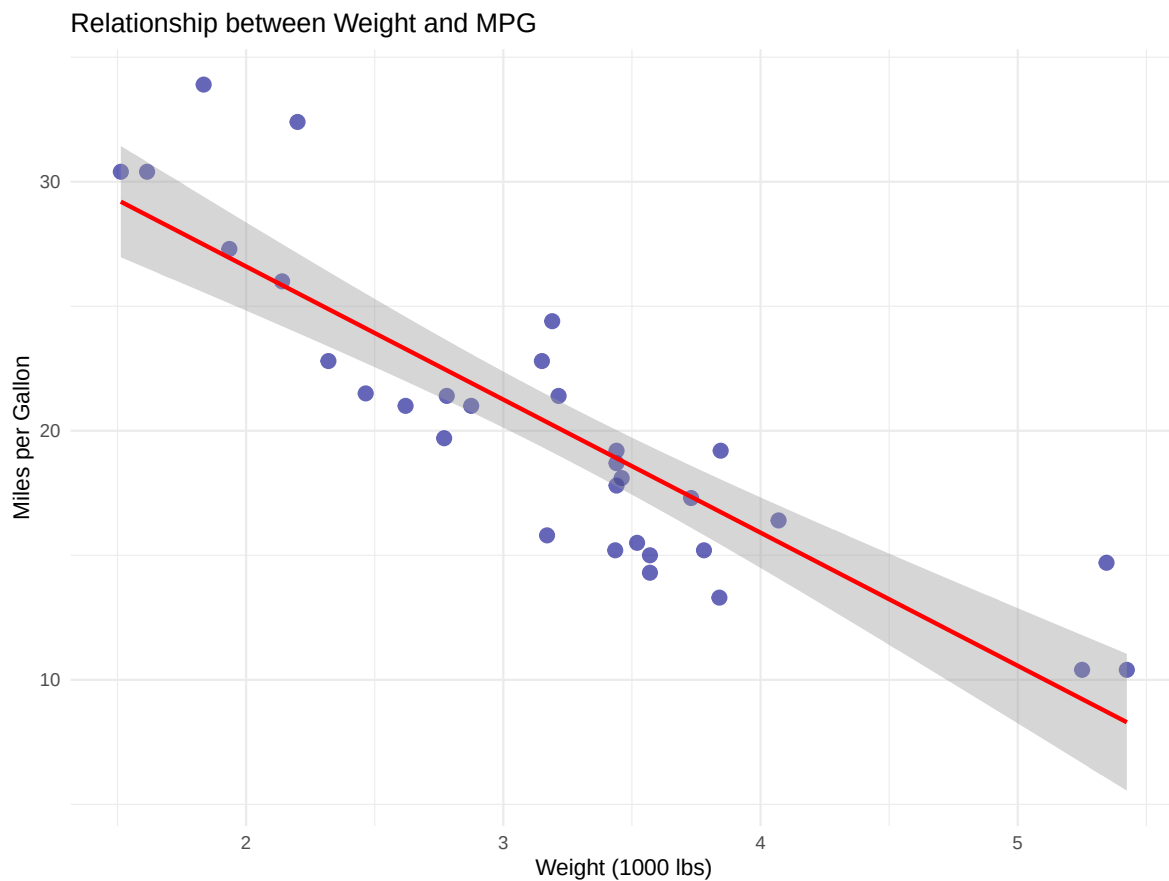


Figure 3: Relationship between variables

4.2.2 Correlation Analysis

```
# Correlation matrix
cor_matrix <- cor(data[, sapply(data, is.numeric)])

# Display correlation table
cor_matrix %>%
  kable(digits = 3, caption = "Correlation matrix")
```

Table 2: Correlation matrix

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
mpg	1.000	-	-	-	0.681	-	0.419	0.664	0.600	0.480	-
		0.852	0.848	0.776		0.868					0.551
cyl	-	1.000	0.902	0.832	-	0.782	-	-	-	-	0.527
	0.852				0.700		0.591	0.811	0.523	0.493	
disp	-	0.902	1.000	0.791	-	0.888	-	-	-	-	0.395
	0.848				0.710		0.434	0.710	0.591	0.556	
hp	-	0.832	0.791	1.000	-	0.659	-	-	-	-	0.750
	0.776				0.449		0.708	0.723	0.243	0.126	
drat	0.681	-	-	-	1.000	-	0.091	0.440	0.713	0.700	-
		0.700	0.710	0.449		0.712					0.091
wt	-	0.782	0.888	0.659	-	1.000	-	-	-	-	0.428
	0.868				0.712		0.175	0.555	0.692	0.583	
qsec	0.419	-	-	-	0.091	-	1.000	0.745	-	-	-
		0.591	0.434	0.708		0.175			0.230	0.213	0.656
vs	0.664	-	-	-	0.440	-	0.745	1.000	0.168	0.206	-
		0.811	0.710	0.723		0.555					0.570
am	0.600	-	-	-	0.713	-	-	0.168	1.000	0.794	0.058
		0.523	0.591	0.243		0.692	0.230				
gear	0.480	-	-	-	0.700	-	-	0.206	0.794	1.000	0.274
		0.493	0.556	0.126		0.583	0.213				
carb	-	0.527	0.395	0.750	-	0.428	-	-	0.058	0.274	1.000
	0.551				0.091		0.656	0.570			

Correlation matrix

5 Statistical Methods

5.1 Methodology

Describe the statistical methods and techniques used in this analysis:

- Method 1: Description and justification
- Method 2: Description and justification

5.2 Assumptions

List and verify any assumptions required for the statistical methods:

1. Assumption 1
2. Assumption 2

5.2.1 Checking Assumptions

```
# Example: Check normality
# qqnorm(data$variable)
# qqline(data$variable)
```

6 Results

6.1 Primary Analysis

6.1.1 Model Fitting

```
# Example: Linear regression
model <- lm(mpg ~ wt + hp + cyl, data = data)

# Model summary
summary(model)
```



```

Call:
lm(formula = mpg ~ wt + hp + cyl, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-3.9290 -1.5598 -0.5311  1.1850  5.8986

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  38.75179     1.78686   21.687 < 2e-16 ***
wt           -3.16697     0.74058   -4.276 0.000199 ***
hp            -0.01804     0.01188   -1.519 0.140015
cyl           -0.94162     0.55092   -1.709 0.098480 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.512 on 28 degrees of freedom
Multiple R-squared:  0.8431,    Adjusted R-squared:  0.8263
F-statistic: 50.17 on 3 and 28 DF,  p-value: 2.184e-11

```

6.1.2 Model Diagnostics

```

# Diagnostic plots
par(mfrow = c(2, 2))
plot(model)

```

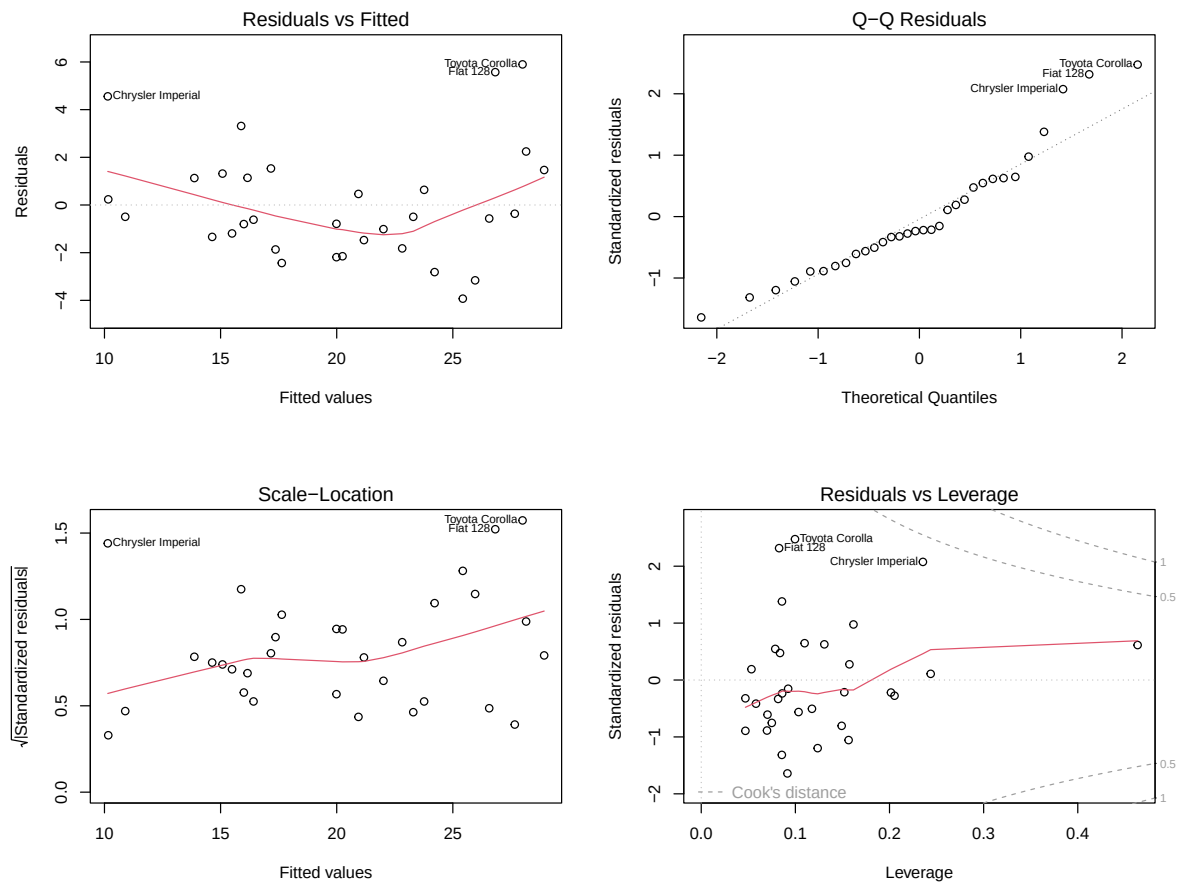


Figure 4: Model diagnostic plots

6.1.3 Results Summary

```
# Extract model coefficients
coef_table <- broom::tidy(model, conf.int = TRUE)

coef_table %>%
  kable(digits = 3, caption = "Model coefficients with 95% confidence intervals")
```

Table 3: Model coefficients with 95% confidence intervals

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	38.752	1.787	21.687	0.000	35.092	42.412
wt	-3.167	0.741	-4.276	0.000	-4.684	-1.650
hp	-0.018	0.012	-1.519	0.140	-0.042	0.006
cyl	-0.942	0.551	-1.709	0.098	-2.070	0.187

6.1.4 Interpretation

Interpret the results in context:

- **Variable 1:** Interpretation
- **Variable 2:** Interpretation

6.2 Secondary Analyses

6.2.1 Additional Tests

```
# Additional statistical tests
# Example: ANOVA
# anova_result <- aov(mpg ~ factor(cyl), data = data)
# summary(anova_result)
```

6.2.2 Subgroup Analysis

If applicable, present any subgroup analyses:

```
# Example: Analysis by subgroup
# data %>%
#   group_by(subgroup) %>%
#   summarise(mean_outcome = mean(outcome),
#             sd_outcome = sd(outcome))
```

6.3 Effect Sizes and Practical Significance

Discuss the practical significance of findings, not just statistical significance:

```
# Calculate effect sizes
# Example: R-squared
r_squared <- summary(model)$r.squared
adj_r_squared <- summary(model)$adj.r.squared

cat("R-squared:", round(r_squared, 3), "\n")
```

R-squared: 0.843

```
cat("Adjusted R-squared:", round(adj_r_squared, 3), "\n")
```

Adjusted R-squared: 0.826

7 Discussion

7.1 Key Findings

Summarize the main findings from the analysis:

1. Finding 1 with interpretation
2. Finding 2 with interpretation
3. Finding 3 with interpretation

7.2 Limitations

Discuss any limitations of the analysis:

- Limitation 1
- Limitation 2
- Limitation 3

7.3 Implications

What are the practical implications of these findings?

7.4 Future Directions

Suggest areas for future research or analysis:

- Direction 1
- Direction 2

8 Conclusions

Provide clear, concise conclusions based on the analysis. Address the original research questions and objectives.

9 References

List any references or citations used in the analysis.

10 Appendix

10.1 Session Information

```
sessionInfo()
```

```
R version 4.5.1 (2025-06-13 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26100)
```

```
Matrix products: default
  LAPACK version 3.12.1
```

```
locale:
[1] LC_COLLATE=English_United Kingdom.utf8
[2] LC_CTYPE=English_United Kingdom.utf8
[3] LC_MONETARY=English_United Kingdom.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United Kingdom.utf8
```

```
time zone: Europe/London
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices  utils      datasets  methods   base
```

```
other attached packages:
```

```
[1] knitr_1.50      lubridate_1.9.4 forcats_1.0.1  stringr_1.5.1  
[5] dplyr_1.1.4     purrr_1.0.4     readr_2.1.5    tidyr_1.3.1  
[9] tibble_3.3.0    ggplot2_3.5.2   tidyverse_2.0.0
```

```
loaded via a namespace (and not attached):
```

```
[1] Matrix_1.7-3      gtable_0.3.6      jsonlite_2.0.0    compiler_4.5.1  
[5] tidyselect_1.2.1  splines_4.5.1     scales_1.4.0      yaml_2.3.10  
[9] fastmap_1.2.0     lattice_0.22-7    R6_2.6.1          labeling_0.4.3  
[13] generics_0.1.4    backports_1.5.0   pillar_1.10.2     RColorBrewer_1.1-3  
[17] tzdb_0.5.0        rlang_1.1.6       broom_1.0.10      stringi_1.8.7  
[21] xfun_0.52         timechange_0.3.0  cli_3.6.5         mgcv_1.9-3  
[25] withr_3.0.2       magrittr_2.0.3    digest_0.6.37     grid_4.5.1  
[29] hms_1.1.3         nlme_3.1-168      lifecycle_1.0.4   vctrs_0.6.5  
[33] evaluate_1.0.4    glue_1.8.0        farver_2.1.2      rmarkdown_2.29  
[37] tools_4.5.1       pkgconfig_2.0.3   htmltools_0.5.8.1
```

10.2 Additional Tables and Figures

Include any supplementary materials here.

```
# Additional code or analyses
```